

Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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Abstract— Movable antennas (MAs) can reshape the spatial correlation between legitimate users and eavesdroppers, which provides a new degree of freedom for physical layer security. However, existing MA-aided secure designs mostly adjust the antenna positions according to instantaneous channel state information (CSI), which is hindered by the mechanical movement delay, the overhead of acquiring CSI over the entire movable region, and the unavailability of the instantaneous CSI of passive eavesdroppers. This paper proposes a two-timescale secure transmission framework for an MA-enabled multiuser downlink in the presence of a passive eavesdropper (Eve). The antenna positions and the long-term power allocation between information signals and artificial noise (AN) are optimized based on the statistical CSI of the legitimate users (LUs) and Eve, while zero-forcing (ZF) precoding and null-space AN are designed based on the instantaneous CSI of the LUs only. By exploiting the conditional distributions induced by ZF precoding and null-space AN, we derive closed-form approximations of the ergodic rates of the LUs and Eve, which depend on the antenna positions only through three geometric quantities characterizing the channel decorrelation among the LUs, the information leakage to Eve, and the AN received by Eve. Based on these expressions, an ergodic secrecy sum rate maximization problem is formulated and solved by an alternating optimization algorithm, in which the power allocation admits a secrecy water-filling structure combined with a one-dimensional search, and the antenna positions are updated element-wise via the majorization-minimization method with analytical gradients and linearized minimum-distance constraints.

Index Terms— Movable antenna, physical layer security, artificial noise, two-timescale design, statistical CSI, ergodic secrecy rate.

I. INTRODUCTION

The rest of this paper is organized as follows. Section II introduces the system model, the two-timescale transmission protocol, and the problem formulation. Section III derives the approximate ergodic secrecy rate. Section IV develops the two-timescale optimization algorithm.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$ denote the transpose, conjugate, and conjugate transpose. $\|\cdot\|$ is the Euclidean norm, $\text{tr}(\cdot)$ the trace, $[\mathbf{X}]_{mn}$ the (m,n) -th entry of $\mathbf{X}$, $[\mathbf{x}]_m$ the m -th entry of $\mathbf{x}$, and $\text{diag}(\cdot)$ a diagonal matrix. $\mathbf{I}_N$ is the $N \times N$ identity matrix and $\mathbf{e}_m$ its m -th column. $\mathcal{CN}(\mu, \Sigma)$ denotes the circularly symmetric complex

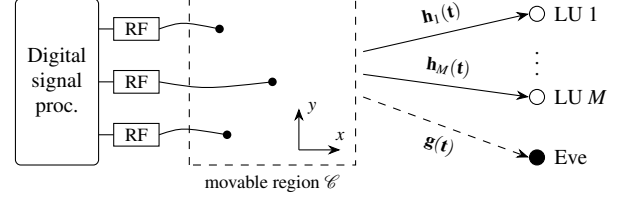


Figure 1: MA-enabled multiuser downlink in the presence of a passive eavesdropper.

Gaussian distribution. $\mathbb{C}^{a \times b}$ denotes the space of $a \times b$ complex matrices, and $\mathbf{X} \succeq \mathbf{Y}$ ($\mathbf{X} \succ \mathbf{Y}$) means that $\mathbf{X} - \mathbf{Y}$ is positive semidefinite (definite). $\mathbb{E}\{\cdot\}$ is the expectation, $\text{Re}\{\cdot\}$ the real part, $[x]^+ = \max\{x, 0\}$, ∇f the gradient of f , and $\mathcal{O}(\cdot)$ the big-O notation.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. MA-Enabled Secure Multiuser Downlink

As shown in Fig. 1, we consider a downlink system in which a base station (BS) equipped with N MAs serves M single-antenna legitimate users (LUs), indexed by $\mathcal{M} \triangleq \{1, \dots, M\}$, in the presence of a single-antenna passive eavesdropper (Eve), where $N > M$. The MAs are connected to the radio-frequency (RF) chains via flexible cables and can move within a two-dimensional (2D) square region $\mathcal{C} \triangleq [-A/2, A/2] \times [-A/2, A/2]$ [1, 2]. The position of the n -th MA is denoted by $\mathbf{t}_n = [x_n, y_n]^T \in \mathcal{C}$, $n \in \mathcal{N} \triangleq \{1, \dots, N\}$, and all positions are collected in $\mathbf{t} \triangleq [\mathbf{t}_1^T, \dots, \mathbf{t}_N^T]^T \in \mathbb{R}^{2N}$. To avoid mutual coupling, any two MAs are separated by at least $D_{\min}$, i.e., $\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}$, $\forall n \neq i$.

Since the size of $\mathcal{C}$ is much smaller than the propagation distances, the far-field condition holds, i.e., moving the MAs changes only the phase of the line-of-sight (LoS) path, while its angle of departure (AoD) and amplitude remain unchanged [2]. Following [3, 4], the channel from the BS to LU m is modeled as Rician fading, i.e.,

$$\mathbf{h}_m(\mathbf{t}) = \sqrt{\frac{\kappa_m \beta_m}{\kappa_m + 1}} \bar{\mathbf{h}}_m(\mathbf{t}) + \sqrt{\frac{\beta_m}{\kappa_m + 1}} \tilde{\mathbf{h}}_m, \quad (1)$$

where β_m is the large-scale fading coefficient, $\kappa_m \geq 0$ is the Rician factor, and $\tilde{\mathbf{h}}_m \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ is the non-LoS (NLoS) com-

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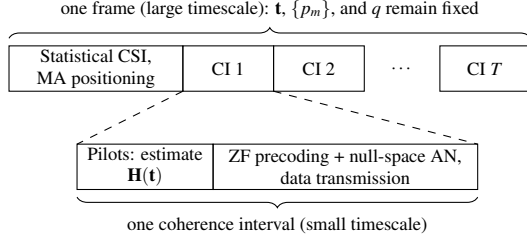


Figure 2: Illustration of the two-timescale transmission protocol (CI: coherence interval).

ponent.¹ The LoS component is given by

$$\bar{\mathbf{h}}_m(\mathbf{t}) = \left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_1^T \mathbf{a}_m}, \dots, e^{j\frac{2\pi}{\lambda}\mathbf{t}_N^T \mathbf{a}_m} \right]^T, \quad (2)$$

where λ is the carrier wavelength, $\mathbf{a}_m \triangleq [\cos \theta_m \sin \phi_m, \sin \theta_m]^T$, and $\theta_m, \phi_m \in [-\pi/2, \pi/2]$ are the elevation and azimuth AoDs of LU m [3]. Similarly, the channel from the BS to Eve is

$$\mathbf{g}(\mathbf{t}) = \sqrt{\frac{\kappa_e \beta_e}{\kappa_e + 1}} \bar{\mathbf{g}}(\mathbf{t}) + \sqrt{\frac{\beta_e}{\kappa_e + 1}} \tilde{\mathbf{g}}, \quad (3)$$

where β_e, κ_e , and $\tilde{\mathbf{g}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ are defined as in (1), and $\bar{\mathbf{g}}(\mathbf{t})$ is given by (2) with $\mathbf{a}_m$ replaced by $\mathbf{a}_e \triangleq [\cos \theta_e \sin \phi_e, \sin \theta_e]^T$. For convenience, we define $\mathbf{H}(\mathbf{t}) \triangleq [\mathbf{h}_1(\mathbf{t}), \dots, \mathbf{h}_M(\mathbf{t})] \in \mathbb{C}^{N \times M}$, $\bar{\mathbf{H}}(\mathbf{t}) \triangleq [\bar{\mathbf{h}}_1(\mathbf{t}), \dots, \bar{\mathbf{h}}_M(\mathbf{t})]$, $\tilde{\mathbf{H}} \triangleq [\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_M]$, and $\mathbf{K} \triangleq \text{diag}(\kappa_1, \dots, \kappa_M)$.

B. Two-Timescale Transmission Protocol

For Rician fading, the NLoS components corresponding to different antenna positions are random, and acquiring the instantaneous CSI over the entire region $\mathcal{C}$ would require the MAs to traverse the region, which incurs prohibitive movement delay and training overhead [3, 4]. Moreover, Eve is passive, so that its instantaneous CSI is unavailable at the BS. In contrast, the statistical CSI, i.e., $\{\beta_m, \kappa_m, \theta_m, \phi_m\}_{m \in \mathcal{M}}$ and $\{\beta_e, \kappa_e, \theta_e, \phi_e\}$, is determined by the large-scale fading and the LoS geometry, which vary slowly and can be acquired with low overhead [3]. Following [5], we assume that the BS knows the statistical CSI of Eve, e.g., by estimating its location, but has no knowledge of the instantaneous NLoS component of the wiretap channel.

Accordingly, we adopt the two-timescale protocol illustrated in Fig. 2. At the beginning of each frame, which consists of T coherence intervals (CIs), the BS optimizes the antenna positions $\mathbf{t}$ and the long-term power allocation based on the statistical CSI and moves the MAs accordingly. Within each CI, with $\mathbf{t}$ fixed, the BS estimates $\mathbf{H}(\mathbf{t})$ via uplink pilots under time-division duplexing and designs the ZF precoders and the AN. We assume that $\mathbf{H}(\mathbf{t})$ is perfectly estimated in each CI, whereas the instantaneous CSI of Eve is never available.

¹ As in [3, 4], the NLoS components at different MAs are assumed to be independent and identically distributed (i.i.d.), which is reasonable in rich-scattering environments with $D_{\min} \geq \lambda/2$. Spatially correlated NLoS components can be incorporated at the cost of more involved expressions.

C. Short-Timescale ZF Precoding and Artificial Noise

Given $\mathbf{t}$, the signal transmitted by the BS in each CI is

$$\mathbf{x} = \sum_{m=1}^M \sqrt{p_m} \mathbf{w}_m s_m + \mathbf{z}, \quad (4)$$

where $s_m \sim \mathcal{CN}(0, 1)$ is the confidential symbol for LU m , $p_m \geq 0$ is its transmit power, $\mathbf{w}_m \in \mathbb{C}^{N \times 1}$ with $\|\mathbf{w}_m\| = 1$ is the ZF beamforming vector, and $\mathbf{z}$ is the AN. The ZF beamformer is given by [3]

$$\mathbf{w}_m = \frac{\mathbf{P}_m^\perp(\mathbf{t}) \mathbf{h}_m(\mathbf{t})}{\|\mathbf{P}_m^\perp(\mathbf{t}) \mathbf{h}_m(\mathbf{t})\|}, \quad \mathbf{P}_m^\perp \triangleq \mathbf{I}_N - \mathbf{B}_m (\mathbf{B}_m^H \mathbf{B}_m)^{-1} \mathbf{B}_m^H, \quad (5)$$

where $\mathbf{B}_m(\mathbf{t}) \in \mathbb{C}^{N \times (M-1)}$ is obtained by removing the m -th column of $\mathbf{H}(\mathbf{t})$, i.e., it collects the channels of the other LUs. The AN is isotropically distributed in the null space of $\mathbf{H}^H(\mathbf{t})$ [6], i.e.,

$$\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, q \mathbf{U}(\mathbf{t})), \quad \mathbf{U}(\mathbf{t}) \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H, \quad (6)$$

where $q \geq 0$ is the AN power per dimension. Since $\mathbf{U}$ is an orthogonal projection matrix of rank $N - M$, the total AN power is $q \text{tr}(\mathbf{U}) = (N - M)q$, and the transmit power constraint is

$$\sum_{m=1}^M p_m + (N - M)q \leq P_{\text{tot}}. \quad (7)$$

Note that the long-term variables $\{p_m\}$ and q are fixed within a frame, whereas $\mathbf{w}_m$ and $\mathbf{U}$ are updated in each CI.

Since $\mathbf{h}_m^H \mathbf{w}_i = 0, \forall i \neq m$, and $\mathbf{h}_m^H \mathbf{U} = \mathbf{0}$, the signal received at LU m is

$$y_m = \mathbf{h}_m^H \mathbf{x} + n_m = \sqrt{p_m} \mathbf{h}_m^H \mathbf{w}_m s_m + n_m, \quad (8)$$

where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$. Hence, both the inter-user interference and the AN are eliminated, and the signal-to-noise ratio (SNR) of LU m is given by [3]

$$\gamma_m = \frac{p_m \|\mathbf{h}_m^H \mathbf{w}_m\|^2}{\sigma_m^2} = \frac{p_m \|\mathbf{P}_m^\perp(\mathbf{t}) \mathbf{h}_m(\mathbf{t})\|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2 [(\mathbf{H}^H \mathbf{H})^{-1}]_{mm}}. \quad (9)$$

The signal received at Eve is $y_e = \mathbf{g}^H \mathbf{x} + n_e$ with $n_e \sim \mathcal{CN}(0, \sigma_e^2)$. To guarantee secrecy in the worst case, we assume that Eve is able to cancel the signals of the other LUs when intercepting s_m [6]. Then, the signal-to-interference-plus-noise ratio (SINR) of Eve for s_m is

$$\gamma_{e,m} = \frac{p_m \|\mathbf{g}^H \mathbf{w}_m\|^2}{q \mathbf{g}^H \mathbf{U} \mathbf{g} + \sigma_e^2}. \quad (10)$$

D. Ergodic Secrecy Rate and Problem Formulation

Since $\mathbf{t}, \{p_m\}$, and q remain fixed over a frame spanning a large number of CIs and Eve's instantaneous CSI is unknown, the confidential messages are encoded over many CIs. By coding over the fading channel variations, the following ergodic secrecy rate of LU m is achievable [6, 7]:

$$R_m^{\text{sec}}(\mathbf{t}, p_m, q) = [\bar{R}_m(\mathbf{t}, p_m) - \bar{R}_{e,m}(\mathbf{t}, p_m, q)]^+, \quad (11)$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1 + \gamma_m)\}, \quad \bar{R}_{e,m} \triangleq \mathbb{E}\{\log_2(1 + \gamma_{e,m})\}, \quad (12)$$

and the expectations are taken over the NLoS components $\tilde{\mathbf{h}}$ and $\tilde{\mathbf{g}}$. We aim to maximize the ergodic secrecy sum rate by jointly optimizing the antenna positions $\mathbf{t}$ and the long-term power allocation $\mathbf{p} \triangleq [p_1, \dots, p_M]^T$ and q , i.e.,

$$(P1) : \max_{\mathbf{t}, \mathbf{p}, q} \sum_{m=1}^M R_m^{\text{sec}}(\mathbf{t}, p_m, q) \quad (13a)$$

$$\text{s.t.} \quad \sum_{m=1}^M p_m + (N-M)q \leq P_{\text{tot}}, \quad (13b)$$

$$p_m \geq 0, \forall m \in \mathcal{M}, \quad q \geq 0, \quad (13c)$$

$$\mathbf{t}_n \in \mathcal{C}, \forall n \in \mathcal{N}, \quad (13d)$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}, \forall n \neq i. \quad (13e)$$

Problem (P1) is challenging since 1) the ergodic rates in (12) have no closed-form expressions; 2) γ_m and $\gamma_{e,m}$ depend on $\mathbf{t}$ through the random projection matrices $\mathbf{P}_m^\perp$ and $\mathbf{U}$; 3) $\mathbf{t}$ is coupled with $\mathbf{p}$ and q in the objective; and 4) constraint (13e) is non-convex. In the following, we first derive tractable approximations of $\bar{R}_m$ and $\bar{R}_{e,m}$ in Section III and then solve the resulting problem in Section IV.

III. ERGODIC SECRECY RATE ANALYSIS

A. Ergodic Rate of the LUs

Conditioned on $\mathbf{B}_m$, i.e., on the channels of the other LUs, $\mathbf{P}_m^\perp$ is a deterministic orthogonal projection matrix of rank $N-M+1$ (with probability one), and $\tilde{\mathbf{h}}_m$ is independent of $\mathbf{B}_m$. Therefore,

$$\mathbf{P}_m^\perp \tilde{\mathbf{h}}_m | \mathbf{B}_m \sim \mathcal{CN}\left(\sqrt{\frac{\kappa_m \beta_m}{\kappa_m + 1}} \mathbf{P}_m^\perp \tilde{\mathbf{h}}_m, \frac{\beta_m}{\kappa_m + 1} \mathbf{P}_m^\perp\right). \quad (14)$$

Let $\mathbf{Q}_m \in \mathbb{C}^{N \times (N-M+1)}$ be a semi-unitary matrix whose columns form an orthonormal basis of the range of $\mathbf{P}_m^\perp$, i.e., $\mathbf{P}_m^\perp = \mathbf{Q}_m \mathbf{Q}_m^H$. Then, $\|\mathbf{P}_m^\perp \tilde{\mathbf{h}}_m\|^2 = \|\mathbf{Q}_m^H \tilde{\mathbf{h}}_m\|^2 = \frac{\beta_m}{\kappa_m + 1} X_m$, where

$$X_m \triangleq \|\sqrt{\kappa_m} \mathbf{Q}_m^H \tilde{\mathbf{h}}_m + \mathbf{Q}_m^H \tilde{\mathbf{h}}_m\|^2 \quad (15)$$

is, conditioned on $\mathbf{B}_m$, the sum of the squared magnitudes of $N-M+1$ independent unit-variance complex Gaussian variables whose squared means sum to $\kappa_m \epsilon_m$, with

$$\epsilon_m \triangleq \|\mathbf{P}_m^\perp \tilde{\mathbf{h}}_m\|^2. \quad (16)$$

In other words, $2X_m$ follows the noncentral chi-square distribution with $2(N-M+1)$ degrees of freedom and noncentrality parameter $2\kappa_m \epsilon_m$, whose moment generating function (MGF) is $\mathbb{E}\{e^{-sX_m} | \mathbf{B}_m\} = (1+s)^{-(N-M+1)} e^{-s\kappa_m \epsilon_m / (1+s)}$ [8]. Accordingly, from (9), $\gamma_m = \zeta_m p_m X_m$ with $\zeta_m \triangleq \beta_m / ((\kappa_m + 1)\sigma_m^2)$.

Since $\log_2(1 + \zeta_m p_m / x)$ is convex with respect to (w.r.t.) $x > 0$, applying Jensen's inequality to X_m^{-1} yields

$$\mathbb{E}\{\log_2(1 + \gamma_m) | \mathbf{B}_m\} \geq \log_2\left(1 + \frac{\zeta_m p_m}{\mathbb{E}\{X_m^{-1} | \mathbf{B}_m\}}\right). \quad (17)$$

Using $1/x = \int_0^\infty e^{-sx} ds$ and the MGF of X_m , we have

$$\begin{aligned} \mathbb{E}\{X_m^{-1} | \mathbf{B}_m\} &= \int_0^\infty (1+s)^{-(N-M+1)} e^{-\frac{s\kappa_m \epsilon_m}{1+s}} ds \\ &\stackrel{(a)}{=} \int_0^1 (1-u)^{N-M-1} e^{-\kappa_m \epsilon_m u} du \\ &\stackrel{(b)}{=} \frac{{}_1F_1(1; N-M+1; -\kappa_m \epsilon_m)}{N-M}, \end{aligned} \quad (18)$$

where (a) follows from the change of variable $u = s/(1+s)$, and (b) follows from the integral representation of the confluent hypergeometric function ${}_1F_1(\cdot; \cdot; \cdot)$ [9, Eq. (13.2.1)]. To obtain a tractable expression, we approximate the inverse moment as

$$\frac{1}{\mathbb{E}\{X_m^{-1} | \mathbf{B}_m\}} \approx \mathbb{E}\{X_m | \mathbf{B}_m\} - 1 = N-M + \kappa_m \epsilon_m, \quad (19)$$

which is exact for $\kappa_m \epsilon_m = 0$ (in which case (18) equals $1/(N-M)$), and whose relative error vanishes as $\kappa_m \epsilon_m$ grows. Substituting (19) into (17) yields

$$\mathbb{E}\{\log_2(1 + \gamma_m) | \mathbf{B}_m\} \approx \log_2(1 + \zeta_m p_m (N-M + \kappa_m \epsilon_m)). \quad (20)$$

For $\kappa_m = 0$, (20) reduces to the well-known lower bound $\log_2(1 + p_m \beta_m (N-M)/\sigma_m^2)$ of the ergodic rate with ZF precoding in Rayleigh fading [10].

The quantity ϵ_m in (20) depends on the NLoS components of the other LUs through $\mathbf{P}_m^\perp$, and is thus random. Next, we approximate it by a deterministic function of $\mathbf{t}$ that depends only on the statistical CSI. To this end, we first consider a generic setup. Let $\mathbf{F} = \mathbf{A} + \mathbf{W} \in \mathbb{C}^{N \times k}$ with $k < N$, where $\mathbf{A}$ is deterministic and $\mathbf{W}$ has i.i.d. $\mathcal{CN}(0, 1)$ entries, and let $\mathbf{f} \in \mathbb{C}^{N \times 1}$ be deterministic. Define $\Phi_{\mathbf{A}} \triangleq \mathbb{E}\{\mathbf{F}^H \mathbf{F}\} = \mathbf{A}^H \mathbf{A} + N\mathbf{I}_k$. By replacing $(\mathbf{F}^H \mathbf{F})^{-1}$ in the projection matrix $\mathbf{P}_{\mathbf{F}} = \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{F}^H$ by $(\mathbb{E}\{\mathbf{F}^H \mathbf{F}\})^{-1} = \Phi_{\mathbf{A}}^{-1}$, which is accurate when either the LoS components dominate or N is large compared with k , and using $\mathbb{E}\{\mathbf{W} \Phi_{\mathbf{A}}^{-1} \mathbf{W}^H\} = \text{tr}(\Phi_{\mathbf{A}}^{-1}) \mathbf{I}_N$, the average projection matrix is approximated by

$$\mathbb{E}\{\mathbf{P}_{\mathbf{F}}\} \approx \underbrace{\mathbf{A} \Phi_{\mathbf{A}}^{-1} \mathbf{A}^H}_{\text{LoS-driven}} + \underbrace{\text{tr}(\Phi_{\mathbf{A}}^{-1}) \mathbf{I}_N}_{\text{NLoS-driven}}. \quad (21)$$

Since $\text{tr}(\mathbf{A} \Phi_{\mathbf{A}}^{-1} \mathbf{A}^H) = \text{tr}(\Phi_{\mathbf{A}}^{-1} (\Phi_{\mathbf{A}} - N\mathbf{I}_k)) = k - N\text{tr}(\Phi_{\mathbf{A}}^{-1})$, the trace of (21) equals $k = \text{tr}(\mathbf{P}_{\mathbf{F}})$, and the NLoS-driven part occupies on average a fraction $\text{tr}(\Phi_{\mathbf{A}}^{-1})$ of each direction of the N -dimensional space. Directly substituting (21) into $\mathbb{E}\{\|\mathbf{P}_{\mathbf{F}}^\perp \mathbf{f}\|^2\} = \|\mathbf{f}\|^2 - \mathbf{f}^H \mathbb{E}\{\mathbf{P}_{\mathbf{F}}\} \mathbf{f}$ counts the energy of $\mathbf{f}$ captured by the LoS-driven part twice, and the result may even become negative when $\mathbf{f}$ is strongly aligned with the columns of $\mathbf{A}$. To avoid this double counting, we apply the fraction $\text{tr}(\Phi_{\mathbf{A}}^{-1})$ only to the LoS residual of $\mathbf{f}$, which leads to

$$\mathbb{E}\{\|\mathbf{P}_{\mathbf{F}}^\perp \mathbf{f}\|^2\} \approx (\|\mathbf{f}\|^2 - \mathbf{f}^H \mathbf{A} \Phi_{\mathbf{A}}^{-1} \mathbf{A}^H \mathbf{f}) (1 - \text{tr}(\Phi_{\mathbf{A}}^{-1})), \quad (22)$$

where $\mathbf{P}_{\mathbf{F}}^\perp = \mathbf{I}_N - \mathbf{P}_{\mathbf{F}}$. Since $\mathbf{0} \preceq \mathbf{A} \Phi_{\mathbf{A}}^{-1} \mathbf{A}^H \prec \mathbf{I}_N$ and $\text{tr}(\Phi_{\mathbf{A}}^{-1}) \leq k/N < 1$, the right-hand side of (22) always lies in $[0, \|\mathbf{f}\|^2]$, as does its left-hand side. Moreover, (22) is exact in the two

extreme cases. For $\mathbf{A} = \mathbf{0}$, $\mathbf{P}_F$ is isotropically distributed with $\mathbb{E}\{\mathbf{P}_F\} = (k/N)\mathbf{I}_N$, and both sides of (22) equal $\|\mathbf{f}\|^2(1 - k/N)$. For $\mathbf{A} = \rho\mathbf{A}_0$, where $\mathbf{A}_0$ has full column rank, both sides converge to $\|(\mathbf{I}_N - \mathbf{A}_0(\mathbf{A}_0^H\mathbf{A}_0)^{-1}\mathbf{A}_0^H)\mathbf{f}\|^2$ as $\rho \rightarrow \infty$.

Since $\mathbf{P}_m^\perp$ is invariant to the scaling of the columns of $\mathbf{B}_m$, we divide the j -th column of $\mathbf{B}_m$ by $\sqrt{\beta_j/(\kappa_j + 1)}$, which yields $\mathbf{B}'_m = \mathbf{A}_m + \tilde{\mathbf{H}}_{-m}$ with

$$\mathbf{A}_m \triangleq \tilde{\mathbf{H}}_{-m}(\mathbf{t})\mathbf{K}_{-m}^{1/2}, \quad (23)$$

where $\tilde{\mathbf{H}}_{-m}$, $\tilde{\mathbf{H}}_{-m}$, and $\mathbf{K}_{-m}$ are obtained by removing the m -th column of $\tilde{\mathbf{H}}$, the m -th column of $\tilde{\mathbf{H}}$, and the m -th row and column of $\mathbf{K}$, respectively. Applying (22) with $\mathbf{f} = \tilde{\mathbf{h}}_m$ ($\|\tilde{\mathbf{h}}_m\|^2 = N$) and $\mathbf{A} = \mathbf{A}_m$ ($k = M - 1$), ε_m is approximated by

$$\tilde{\varepsilon}_m(\mathbf{t}) \triangleq (N - \tilde{\mathbf{h}}_m^H \mathbf{A}_m \Phi_m^{-1} \mathbf{A}_m^H \tilde{\mathbf{h}}_m)(1 - \text{tr}(\Phi_m^{-1})), \quad (24)$$

where $\Phi_m \triangleq \mathbf{A}_m^H \mathbf{A}_m + N\mathbf{I}_{M-1}$.

Evaluating (24) for all LUs requires M matrix inversions of size $M - 1$. A more compact expression can be obtained by defining

$$\Phi(\mathbf{t}) \triangleq \mathbf{K}^{1/2} \tilde{\mathbf{H}}^H(\mathbf{t}) \tilde{\mathbf{H}}(\mathbf{t}) \mathbf{K}^{1/2} + N\mathbf{I}_M, \quad \Omega(\mathbf{t}) \triangleq \Phi^{-1}(\mathbf{t}). \quad (25)$$

Note that Φ_m is the submatrix of Φ obtained by removing its m -th row and column, the m -th column of Φ without its m -th entry is $\sqrt{\kappa_m} \mathbf{A}_m^H \tilde{\mathbf{h}}_m$, and $[\Phi]_{mm} = N(\kappa_m + 1)$. By the block matrix inversion formula [12], we have

$$[\Omega]_{mm} = \frac{1}{N(\kappa_m + 1) - \kappa_m \tilde{\mathbf{h}}_m^H \mathbf{A}_m \Phi_m^{-1} \mathbf{A}_m^H \tilde{\mathbf{h}}_m}, \quad (26)$$

$$\Phi_m^{-1} = [\Omega]_{-m,-m} - \frac{[\Omega]_{-m,m}[\Omega]_{m,-m}}{[\Omega]_{mm}}, \quad (27)$$

where $[\Omega]_{-m,-m}$ is obtained by removing the m -th row and column of Ω , and $[\Omega]_{-m,m}$ ($[\Omega]_{m,-m}$) is the m -th column (row) of Ω without its m -th entry. From (26), $\kappa_m(N - \tilde{\mathbf{h}}_m^H \mathbf{A}_m \Phi_m^{-1} \mathbf{A}_m^H \tilde{\mathbf{h}}_m) = 1/[\Omega]_{mm} - N$. Taking the trace of (27) and using $[\Omega^2]_{mm} = \sum_j |[\Omega]_{jm}|^2$, we obtain $\text{tr}(\Phi_m^{-1}) = \text{tr}(\Omega) - [\Omega^2]_{mm}/[\Omega]_{mm}$. Hence, the effective LoS energy of LU m after ZF, $\mu_m(\mathbf{t}) \triangleq \kappa_m \tilde{\varepsilon}_m(\mathbf{t})$, can be expressed as

$$\mu_m(\mathbf{t}) = \left(\frac{1}{[\Omega(\mathbf{t})]_{mm}} - N \right) \left(1 - \text{tr}(\Omega(\mathbf{t})) + \frac{[\Omega^2(\mathbf{t})]_{mm}}{[\Omega(\mathbf{t})]_{mm}} \right). \quad (28)$$

Since $\Phi \succeq N\mathbf{I}_M$ and $\Phi_m \succeq N\mathbf{I}_{M-1}$, we have $[\Omega]_{mm} \leq 1/N$ and $\text{tr}(\Phi_m^{-1}) \leq (M - 1)/N < 1$, and thus $\mu_m(\mathbf{t}) \geq 0$. Moreover, since $\tilde{\mathbf{h}}_m^H \mathbf{A}_m \Phi_m^{-1} \mathbf{A}_m^H \tilde{\mathbf{h}}_m \geq 0$, (26) yields $1/[\Omega]_{mm} - N \leq \kappa_m N$, and hence $\mu_m(\mathbf{t}) \leq \kappa_m N$. Finally, by replacing $\kappa_m \varepsilon_m$ in (20) by its approximate mean $\mu_m(\mathbf{t})$, which is reasonable since ε_m concentrates around its mean in the two regimes discussed above, the ergodic rate of LU m is approximated by

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t}, p_m) \triangleq \log_2(1 + \zeta_m p_m(N - M + \mu_m(\mathbf{t}))). \quad (29)$$

Remark 1 (Accuracy of (29)): The matrix $\Sigma(\mathbf{t})$ in [3, Eq. (38)] equals $\frac{1}{N}(\mathbf{K} + \mathbf{I}_M)^{-1/2} \Phi(\mathbf{t})(\mathbf{K} + \mathbf{I}_M)^{-1/2}$. Hence,

the lower bound on the ergodic rate with ZF precoding in [3, Eq. (37)] can be rewritten in our notation as $\log_2(1 + \zeta_m p_m(N - M)/(N[\Omega]_{mm}))$. Using (28) and $\text{tr}(\Phi_m^{-1}) = \text{tr}(\Omega) - [\Omega^2]_{mm}/[\Omega]_{mm}$, the difference between the effective SNR terms of (29) and this bound is

$$N - M + \mu_m - \frac{N - M}{N[\Omega]_{mm}} = \left(\frac{1}{[\Omega]_{mm}} - N \right) \left(\frac{M}{N} - \text{tr}(\Phi_m^{-1}) \right) \geq 0, \quad (30)$$

where the equality holds if and only if $\kappa_m = 0$. Moreover, as the Rician factors grow, both $\gamma_m[\Omega]_{mm}/(\zeta_m p_m)$ and $(N - M + \mu_m)/[\Omega]_{mm}$ tend to one, whereas the corresponding quantity of the bound equals $(N - M)/N < 1$. Hence, (29) captures the LoS-induced SNR gain, which is exactly the part that can be reshaped by the MAs, more accurately.

B. Ergodic Wiretap Rate at Eve

The numerator and the denominator of $\gamma_{e,m}$ in (10) are the information leakage $p_m |\mathbf{g}^H \mathbf{w}_m|^2$ and the AN-plus-noise power $q \mathbf{g}^H \mathbf{U} \mathbf{g} + \sigma_e^2$, respectively. Given $\mathbf{H}$, both $\mathbf{w}_m$ and $\mathbf{U}$ are fixed and independent of $\tilde{\mathbf{g}}$. Hence, from (3), $\|\mathbf{w}_m\| = 1$, and $\text{tr}(\mathbf{U}) = N - M$, we have

$$\mathbb{E}\{|\mathbf{g}^H \mathbf{w}_m|^2 | \mathbf{H}\} = \tilde{\beta}_e (\kappa_e |\tilde{\mathbf{g}}^H \mathbf{w}_m|^2 + 1), \quad (31)$$

$$\mathbb{E}\{\mathbf{g}^H \mathbf{U} \mathbf{g} | \mathbf{H}\} = \tilde{\beta}_e (\kappa_e \tilde{\mathbf{g}}^H \mathbf{U} \tilde{\mathbf{g}} + N - M), \quad (32)$$

where $\tilde{\beta}_e \triangleq \beta_e/(\kappa_e + 1)$. Since both the numerator and the denominator of $\gamma_{e,m}$ are nonnegative random variables, we apply [11, Lemma 1], i.e., $\mathbb{E}\{\log_2(1 + X/Y)\} \approx \log_2(1 + \mathbb{E}\{X\}/\mathbb{E}\{Y\})$, and obtain from (31) and (32) that

$$\bar{R}_{e,m} \approx \log_2 \left(1 + \frac{p_m (\kappa_e \mathbb{E}\{|\tilde{\mathbf{g}}^H \mathbf{w}_m|^2\} + 1)}{q (\kappa_e \mathbb{E}\{\tilde{\mathbf{g}}^H \mathbf{U} \tilde{\mathbf{g}}\} + N - M) + \bar{\sigma}_e^2} \right), \quad (33)$$

where $\bar{\sigma}_e^2 \triangleq \sigma_e^2/\tilde{\beta}_e$ and the remaining expectations are taken over $\tilde{\mathbf{H}}$. Given $\mathbf{H}$, the AN power received by Eve is the sum of $N - M$ independent terms, and thus (33) becomes more accurate as the number of AN dimensions $N - M$ grows [11]. It remains to evaluate the two expectations in (33).

1) *Information leakage* $\mathbb{E}\{|\tilde{\mathbf{g}}^H \mathbf{w}_m|^2\}$: The ZF direction $\mathbf{w}_m$ is invariant to the scaling of the columns of $\mathbf{H}$. Dividing the j -th column of $\mathbf{H}$ by $\sqrt{\beta_j/(\kappa_j + 1)}$ yields $\mathbf{H}' \triangleq \tilde{\mathbf{H}} \mathbf{K}^{1/2} + \tilde{\mathbf{H}}$, and $\mathbf{w}_m$ can be equivalently expressed as the normalized m -th column of the ZF precoding matrix $\mathbf{H}'(\mathbf{H}'^H \mathbf{H}')^{-1}$, i.e.,

$$\mathbf{w}_m = \frac{\mathbf{H}'(\mathbf{H}'^H \mathbf{H}')^{-1} \mathbf{e}_m}{\|\mathbf{H}'(\mathbf{H}'^H \mathbf{H}')^{-1} \mathbf{e}_m\|}. \quad (34)$$

By replacing $(\mathbf{H}'^H \mathbf{H}')^{-1}$ in (34) by $(\mathbb{E}\{\mathbf{H}'^H \mathbf{H}'\})^{-1} = \Omega$, we have $|\tilde{\mathbf{g}}^H \mathbf{w}_m|^2 \approx |\tilde{\mathbf{g}}^H \mathbf{H}' \Omega \mathbf{e}_m|^2 / \|\mathbf{H}' \Omega \mathbf{e}_m\|^2$. Then, approximating the expectation of this ratio by the ratio of the expectations of its numerator and denominator, in the same spirit as [11, Lemma 1], and using

$$\mathbb{E}\{\mathbf{H}'^H \tilde{\mathbf{g}} \tilde{\mathbf{g}}^H \mathbf{H}'\} = \mathbf{v} \mathbf{v}^H + N\mathbf{I}_M, \quad \mathbf{v}(\mathbf{t}) \triangleq \mathbf{K}^{1/2} \tilde{\mathbf{H}}^H(\mathbf{t}) \tilde{\mathbf{g}}(\mathbf{t}), \quad (35)$$

$$\mathbb{E}\{\|\mathbf{H}' \Omega \mathbf{e}_m\|^2\} = [\Omega \Phi \Omega]_{mm} = [\Omega]_{mm}, \quad (36)$$

we obtain

$$\mathbb{E}\{|\bar{\mathbf{g}}^H \mathbf{w}_m|^2\} \approx \bar{c}_m(\mathbf{t}) \triangleq \frac{|\left[\Omega(\mathbf{t})\mathbf{v}(\mathbf{t})\right]_m|^2 + N[\Omega^2(\mathbf{t})]_{mm}}{[\Omega(\mathbf{t})]_{mm}}. \quad (37)$$

2) *AN at Eve* $\mathbb{E}\{\bar{\mathbf{g}}^H \mathbf{U} \bar{\mathbf{g}}\}$: Since $\mathbf{U}$ is also invariant to the column scaling of $\mathbf{H}$, $\bar{\mathbf{g}}^H \mathbf{U} \bar{\mathbf{g}} = \|\mathbf{P}_{\bar{\mathbf{H}}}^\perp \bar{\mathbf{g}}\|^2$. Applying (22) with $\mathbf{f} = \bar{\mathbf{g}}$ and $\mathbf{A} = \bar{\mathbf{H}}\mathbf{K}^{1/2}$ ($k = M$), for which $\Phi_{\mathbf{A}} = \Phi$ and $\mathbf{A}^H \mathbf{f} = \mathbf{v}$, we obtain

$$\mathbb{E}\{\bar{\mathbf{g}}^H \mathbf{U} \bar{\mathbf{g}}\} \approx \bar{d}(\mathbf{t}) \triangleq (N - \mathbf{v}^H(\mathbf{t})\Omega(\mathbf{t})\mathbf{v}(\mathbf{t}))(1 - \text{tr}(\Omega(\mathbf{t}))). \quad (38)$$

Substituting (37) and (38) into (33), the ergodic wiretap rate of LU m is approximated by

$$\begin{aligned} \bar{R}_{e,m} &\approx \bar{R}_{e,m}(\mathbf{t}, p_m, q) \\ &\triangleq \log_2 \left(1 + \frac{p_m(\kappa_e \bar{c}_m(\mathbf{t}) + 1)}{q(\kappa_e \bar{d}(\mathbf{t}) + N - M) + \bar{\sigma}_e^2} \right). \end{aligned} \quad (39)$$

C. Approximate Ergodic Secrecy Rate and Discussions

Based on (29) and (39), the ergodic secrecy rate of LU m is approximated by

$$R_m^{\text{sec}} \approx \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q) \triangleq [\tilde{R}_m(\mathbf{t}, p_m) - \bar{R}_{e,m}(\mathbf{t}, p_m, q)]^+, \quad (40)$$

which depends only on the statistical CSI and is thus suitable for the large-timescale design.

Remark 2 (Role of the antenna positions): From (28), (37), (38), and (25), the antenna positions affect the secrecy rate only through $\Phi(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$, i.e., through the LoS array correlations

$$[\bar{\mathbf{H}}^H \bar{\mathbf{H}}]_{ij} = \sum_{n=1}^N e^{j\frac{2\pi}{\lambda} \mathbf{t}_n^T (\mathbf{a}_j - \mathbf{a}_i)}, \quad [\bar{\mathbf{H}}^H \bar{\mathbf{g}}]_i = \sum_{n=1}^N e^{j\frac{2\pi}{\lambda} \mathbf{t}_n^T (\mathbf{a}_e - \mathbf{a}_i)}. \quad (41)$$

Specifically, $\mu_m(\mathbf{t})$ reflects the decorrelation between LU m and the other LUs, which determines the ZF gain; $\bar{c}_m(\mathbf{t})$ is the LoS energy of Eve along the ZF beam of LU m , i.e., the information leakage; and $\bar{d}(\mathbf{t})$ is the LoS energy of Eve in the AN subspace, i.e., the AN received by Eve. By optimizing $\mathbf{t}$, the MA array can simultaneously decrease $\bar{c}_m(\mathbf{t})$ and increase $\bar{d}(\mathbf{t})$ by decorrelating the LoS component of Eve from those of the LUs, which is difficult for fixed-position antennas (FPAs) when Eve is spatially close to an LU.

Remark 3 (Special cases): 1) If $\kappa_e = 0$, (39) reduces to $\log_2(1 + p_m/(q(N - M) + \bar{\sigma}_e^2))$, which does not depend on $\mathbf{t}$, i.e., the statistical CSI of Eve carries no spatial information, and the MAs enhance secrecy only through $\mu_m(\mathbf{t})$. 2) If $M = 1$, $\Phi = N(\kappa_1 + 1)$ is a scalar, and thus $\mu_1 = \kappa_1 N$ is independent of $\mathbf{t}$, while

$$\bar{c}_1(\mathbf{t}) = \frac{\kappa_1 |\bar{\mathbf{h}}_1^H(\mathbf{t}) \bar{\mathbf{g}}(\mathbf{t})|^2 / N + 1}{\kappa_1 + 1}, \quad (42)$$

$$\bar{d}(\mathbf{t}) = \left(N - \frac{\kappa_1 |\bar{\mathbf{h}}_1^H(\mathbf{t}) \bar{\mathbf{g}}(\mathbf{t})|^2}{N(\kappa_1 + 1)} \right) \left(1 - \frac{1}{N(\kappa_1 + 1)} \right) \quad (43)$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t}) \bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Hence, different from the rate-oriented two-timescale design where MAs bring no gain for a single user [3,

Remark 2], MAs still enhance the secrecy performance by reducing the LoS correlation between the LU and Eve.

Remark 4 (High-SNR power split): Consider equal power allocation $p_m = \alpha P_{\text{tot}}/M$ and $q = (1 - \alpha)P_{\text{tot}}/(N - M)$ with $\alpha \in (0, 1)$. As $P_{\text{tot}} \rightarrow \infty$, we have

$$\begin{aligned} \bar{R}_m - \bar{R}_{e,m} &= \log_2 \frac{\alpha P_{\text{tot}} \zeta_m (N - M + \mu_m)}{M} \\ &\quad - \log_2 \left(1 + \frac{\alpha K_m}{1 - \alpha} \right) + o(1), \end{aligned} \quad (44)$$

where $K_m \triangleq \frac{N-M}{M} \cdot \frac{\kappa_e \bar{c}_m + 1}{\kappa_e \bar{d} + N - M}$. Hence, the high-SNR slope does not depend on $\mathbf{t}$, and the MA positions affect the secrecy rate through the constant term. The value of α maximizing the sum of (44) over m is the root of $\sum_m \left[\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{K_m-1}{1+\alpha(K_m-1)} \right] = 0$, which, for $K_m = K, \forall m$, is given by

$$\alpha^* = \frac{1}{1 + \sqrt{K}}. \quad (45)$$

Since the MA array can decrease $\bar{c}_m$ and increase $\bar{d}$, it decreases K and increases α^* , i.e., less AN power is needed.

IV. PROPOSED TWO-TIMESCALE OPTIMIZATION ALGORITHM

A. Problem Reformulation

To facilitate the optimization, we define

$$\begin{aligned} \eta_m(\mathbf{t}) &\triangleq \zeta_m (N - M + \mu_m(\mathbf{t})), \quad \xi_m(\mathbf{t}) \triangleq \kappa_e \bar{c}_m(\mathbf{t}) + 1, \\ \psi(\mathbf{t}) &\triangleq \kappa_e \bar{d}(\mathbf{t}) + N - M, \end{aligned} \quad (46)$$

which represent the effective channel gain of LU m , the normalized information leakage of LU m , and the normalized AN power received by Eve, respectively. By replacing the ergodic secrecy rates in (P1) by (40), we consider the following problem:

$$\begin{aligned} (\text{P2}) : \max_{\mathbf{t}, \mathbf{p}, q} F(\mathbf{t}, \mathbf{p}, q) &\triangleq \sum_{m=1}^M \left[\log_2(1 + \eta_m(\mathbf{t}) p_m) \right. \\ &\quad \left. - \log_2 \left(1 + \frac{\xi_m(\mathbf{t}) p_m}{q \psi(\mathbf{t}) + \bar{\sigma}_e^2} \right) \right] \end{aligned} \quad (47a)$$

$$\text{s.t. (13b)–(13e).} \quad (47b)$$

The operator $[\cdot]^+$ is dropped in (P2) without loss of optimality. Specifically, for any feasible point at which the m -th term of F is negative, setting $p_m = 0$ and reallocating this power to the AN, i.e., increasing q , makes the m -th term zero and does not decrease the other terms, which are non-decreasing in q . Hence, at the optimal solution of (P2), all terms of F are non-negative, and (P2) and its counterpart with $[\cdot]^+$ have the same optimal value. Problem (P2) is still non-convex due to the coupling between $\mathbf{t}$ and $(\mathbf{p}, q)$ and the constraint (13e). We adopt the alternating optimization (AO) framework, which optimizes $(\mathbf{p}, q)$ and $\mathbf{t}$ alternately.

B. Power Allocation Optimization

For given $\mathbf{t}$, η_m , ξ_m , and ψ are constants, and (P2) reduces to

$$(P3) : \max_{\mathbf{p}, q} \sum_{m=1}^M f_m(p_m; q) \quad (48a)$$

$$\text{s.t. (13b), (13c),} \quad (48b)$$

where

$$f_m(p; q) \triangleq \log_2(1 + \eta_m p) - \log_2(1 + \vartheta_m(q)p) \quad (49)$$

with $\vartheta_m(q) \triangleq \xi_m/(q\psi + \bar{\sigma}_e^2)$ being the effective channel gain of Eve for LU m , which decreases with q . We first optimize $\mathbf{p}$ for a given q . The first-order and second-order derivatives of f_m w.r.t. p are

$$f'_m(p) = \frac{\eta_m - \vartheta_m}{\ln 2 (1 + \eta_m p)(1 + \vartheta_m p)}, \quad (50)$$

$$f''_m(p) = \frac{1}{\ln 2} \left[\frac{\vartheta_m^2}{(1 + \vartheta_m p)^2} - \frac{\eta_m^2}{(1 + \eta_m p)^2} \right]. \quad (51)$$

If $\eta_m \leq \vartheta_m$, then $f_m(p) \leq 0$, $\forall p \geq 0$, and hence $p_m^* = 0$. If $\eta_m > \vartheta_m$, then $\eta_m(1 + \vartheta_m p) - \vartheta_m(1 + \eta_m p) = \eta_m - \vartheta_m > 0$, i.e., $\frac{\eta_m}{1 + \eta_m p} > \frac{\vartheta_m}{1 + \vartheta_m p}$, which implies $f''_m(p) < 0$; thus, f_m is increasing and concave. Let $\mathcal{M}_a(q) \triangleq \{m \mid \eta_m > \vartheta_m(q)\}$ and $P_1(q) \triangleq P_{\text{tot}} - (N - M)q$. For a given q , the power allocation problem

$$\max_{\{p_m \geq 0\}} \sum_{m \in \mathcal{M}_a(q)} f_m(p_m; q) \quad \text{s.t.} \quad \sum_{m \in \mathcal{M}_a(q)} p_m \leq P_1(q), \quad (52)$$

with $p_m = 0$ for $m \notin \mathcal{M}_a(q)$, is convex, and the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality [13]. Introducing the dual variable $\nu \geq 0$, the stationarity condition for $p_m > 0$ is $f'_m(p_m) = \nu$, i.e.,

$$(1 + \eta_m p_m)(1 + \vartheta_m p_m) = \frac{\eta_m - \vartheta_m}{\nu \ln 2}, \quad (53)$$

which is a quadratic equation in p_m . Its positive root, rewritten in a numerically stable form, together with the complementary slackness condition, yields

$$p_m^*(\nu) = \left[\frac{2 \left(\frac{\eta_m - \vartheta_m}{\nu \ln 2} - 1 \right)}{\sqrt{\Delta_m(\nu)} + \eta_m + \vartheta_m} \right]^+, \quad (54)$$

where $\Delta_m(\nu) \triangleq (\eta_m - \vartheta_m)^2 + 4\eta_m \vartheta_m (\eta_m - \vartheta_m) / (\nu \ln 2)$. Note that $p_m^*(\nu) > 0$ if and only if $\nu < f'_m(0) = (\eta_m - \vartheta_m) / \ln 2$, and that (54) reduces to the classic water-filling solution $[1/(\nu \ln 2) - 1/\eta_m]^+$ when $\vartheta_m \rightarrow 0$. Hence, (54) can be regarded as a secrecy water-filling solution, in which the LUs whose effective channels are weaker than that of Eve are switched off. Since $p_m^*(\nu)$ is non-increasing in ν , the optimal ν satisfying $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu) = P_1(q)$ can be found by the bisection method over $(0, \max_{m \in \mathcal{M}_a(q)} (\eta_m - \vartheta_m) / \ln 2)$.

Next, let $G(q) \triangleq \sum_{m \in \mathcal{M}_a(q)} f_m(p_m^*(\nu^*(q)); q)$ denote the objective value achieved by the optimal power allocation for a

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Require: $\{\eta_m, \xi_m\}_{m \in \mathcal{M}}$, ψ , P_{tot} , and grid $\mathcal{Q}$.

1: **for** each $q \in \mathcal{Q}$ **do**

2: Compute $\vartheta_m(q)$ by (49) and $\mathcal{M}_a(q)$.

3: Find ν by bisection such that $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu) = P_1(q)$ with (54).

4: Evaluate $G(q)$.

5: **end for**

6: Refine q^* around $\arg \max_{q \in \mathcal{Q}} G(q)$ by golden-section search, and compute $\mathbf{p}^*$ by (54).

Ensure: $\mathbf{p}^*$ and q^* .

given q , where $\nu^*(q)$ is obtained by the above bisection, and $G(q) = 0$ if $\mathcal{M}_a(q) = \emptyset$ or $P_1(q) = 0$. The optimal AN power is obtained by the one-dimensional search

$$q^* = \arg \max_{0 \leq q \leq q_{\max}} G(q), \quad (55)$$

where $q_{\max} \triangleq P_{\text{tot}} / (N - M)$, and (55) is performed by a grid search over a set $\mathcal{Q} \subset [0, q_{\max}]$ followed by a golden-section refinement around the best grid point. Increasing q reduces the effective channel gains $\vartheta_m(q)$ of Eve at the cost of less power for the confidential signals, and (55) finds the best tradeoff. The procedure is summarized in Algorithm 1.

C. Antenna Position Optimization

For given $\mathbf{p}$ and q , (P2) reduces to

$$(P4) : \max_{\mathbf{t}} F(\mathbf{t}) \quad \text{s.t. (13d), (13e),} \quad (56)$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$ for brevity. Problem (P4) is non-convex since $F(\mathbf{t})$ is neither concave nor convex w.r.t. $\mathbf{t}$. Similar to [3], we optimize the position of each MA in an element-wise manner, with the positions of the other MAs being fixed.

1) *Dependence on $\mathbf{t}_n$:* Let $\mathbf{b}_n(\mathbf{t}_n) \in \mathbb{C}^{M \times 1}$ denote the conjugate transpose of the n -th row of $\bar{\mathbf{H}}(\mathbf{t})$, i.e.,

$$\mathbf{b}_n(\mathbf{t}_n) = [e^{-j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_1}, \dots, e^{-j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_M}]^T, \quad (57)$$

and let $\mathbf{k}_n(\mathbf{t}_n) \triangleq \mathbf{K}^{1/2} \mathbf{b}_n(\mathbf{t}_n)$. Since $\bar{\mathbf{H}}^H \bar{\mathbf{H}} = \sum_{i=1}^N \mathbf{b}_i \mathbf{b}_i^H$ and $\bar{\mathbf{H}}^H \bar{\mathbf{g}} = \sum_{i=1}^N \mathbf{b}_i e^{j\frac{2\pi}{\lambda} \mathbf{t}_i^T \mathbf{a}_e}$, $\Phi(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$ can be expressed as

$$\Phi(\mathbf{t}) = \Gamma_n + \mathbf{k}_n(\mathbf{t}_n) \mathbf{k}_n^H(\mathbf{t}_n), \quad (58)$$

$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \mathbf{k}_n(\mathbf{t}_n) e^{j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_e}, \quad (59)$$

where

$$\Gamma_n \triangleq \sum_{i \neq n} \mathbf{k}_i \mathbf{k}_i^H + N \mathbf{I}_M, \quad \mathbf{r}_n \triangleq \sum_{i \neq n} \mathbf{k}_i e^{j\frac{2\pi}{\lambda} \mathbf{t}_i^T \mathbf{a}_e} \quad (60)$$

do not depend on $\mathbf{t}_n$, with $\mathbf{k}_i \triangleq \mathbf{k}_i(\mathbf{t}_i)$. By the Sherman–Morrison formula [12],

$$\Omega(\mathbf{t}) = \Gamma_n^{-1} - \frac{\Gamma_n^{-1} \mathbf{k}_n(\mathbf{t}_n) \mathbf{k}_n^H(\mathbf{t}_n) \Gamma_n^{-1}}{1 + \mathbf{k}_n^H(\mathbf{t}_n) \Gamma_n^{-1} \mathbf{k}_n(\mathbf{t}_n)}, \quad (61)$$

so that, for given Γ_n^{-1} and $\mathbf{r}_n$, evaluating μ_m , $\bar{c}_m$, $\bar{d}$, and hence F as functions of $\mathbf{t}_n$ only requires $\mathcal{O}(M^2)$ operations.

2) *Gradient*: For $u \in \{x_n, y_n\}$, let $\mathbf{D}_u \triangleq \text{diag}(a_{1,u}, \dots, a_{M,u})$, where $a_{m,x} = \cos \theta_m \sin \phi_m$ and $a_{m,y} = \sin \theta_m$ are the entries of $\mathbf{a}_m$, and let $a_{e,u}$ denote the corresponding entry of $\mathbf{a}_e$. From (58)–(59), we have

$$\dot{\mathbf{k}}_n \triangleq \frac{\partial \mathbf{k}_n}{\partial u} = -j \frac{2\pi}{\lambda} \mathbf{D}_u \mathbf{k}_n, \quad (62)$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u} = j \frac{2\pi}{\lambda} (a_{e,u} \mathbf{I}_M - \mathbf{D}_u) \mathbf{k}_n e^{j \frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_e}, \quad (63)$$

$$\dot{\Omega} \triangleq \frac{\partial \Omega}{\partial u} = -\Omega (\dot{\mathbf{k}}_n \mathbf{k}_n^H + \mathbf{k}_n \dot{\mathbf{k}}_n^H) \Omega, \quad (64)$$

where (64) follows from $\partial \Phi^{-1} = -\Phi^{-1}(\partial \Phi) \Phi^{-1}$. Based on (62)–(64), the derivatives $\dot{\mu}_m \triangleq \partial \mu_m / \partial u$, $\dot{c}_m \triangleq \partial \bar{c}_m / \partial u$, and $\dot{d} \triangleq \partial \bar{d} / \partial u$ are derived in closed form in Appendix A. Let $\Lambda_m \triangleq \xi_m p_m$ and $\Upsilon \triangleq q\psi + \bar{\sigma}_e^2$, whose derivatives are $\dot{\Lambda}_m = \kappa_e p_m \dot{c}_m$ and $\dot{\Upsilon} = \kappa_e q \dot{d}$. Using $\frac{\partial}{\partial u} \log_2(1 + \Lambda_m / \Upsilon) = \frac{\dot{\Lambda}_m \Upsilon - \Lambda_m \dot{\Upsilon}}{\ln 2 \Upsilon (\Lambda_m + \Upsilon)}$ and the chain rule, the derivative of F w.r.t. u is

$$\frac{\partial F}{\partial u} = \frac{1}{\ln 2} \sum_{m=1}^M \left[\frac{\xi_m p_m \dot{\mu}_m}{1 + \eta_m p_m} - \frac{\kappa_e p_m \dot{c}_m}{\Lambda_m + \Upsilon} + \frac{\kappa_e q \Lambda_m \dot{d}}{\Upsilon (\Lambda_m + \Upsilon)} \right], \quad (65)$$

and the gradient is $\nabla_{\mathbf{t}_n} F = [\partial F / \partial x_n, \partial F / \partial y_n]^T$.

3) *Surrogate function*: With a slight abuse of notation, let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the positions of the other MAs fixed at their latest values. Since F is twice continuously differentiable and $\mathcal{C}$ is convex and compact, $\nabla_{\mathbf{t}_n} F$ is Lipschitz continuous over $\mathcal{C}$ with some constant L_n . According to the descent lemma [14, Lemma 5.7], for any $\delta_n \geq L_n$, the following quadratic function is a lower bound of F that is tight at the local point $\mathbf{t}_n^{(\ell)}$ in the ℓ -th iteration:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n | \mathbf{t}_n^{(\ell)}) \triangleq F(\mathbf{t}_n^{(\ell)}) + \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(\ell)})^T (\mathbf{t}_n - \mathbf{t}_n^{(\ell)}) - \frac{\delta_n}{2} \|\mathbf{t}_n - \mathbf{t}_n^{(\ell)}\|^2. \quad (66)$$

Since L_n is difficult to obtain in closed form, δ_n is determined by the backtracking procedure [14, 15]: starting from an initial value δ_0 , δ_n is doubled until the solution of the subproblem below satisfies $F(\mathbf{t}_n^{(\ell+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)})$, which is guaranteed to occur once $\delta_n \geq L_n$.

4) *Minimum-distance constraints*: Since $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ is convex w.r.t. $\mathbf{t}_n$, it is lower-bounded by its first-order Taylor expansion [3], i.e.,

$$\|\mathbf{t}_n - \mathbf{t}_i\|^2 \geq \|\mathbf{d}_{ni}^{(\ell)}\|^2 + 2(\mathbf{d}_{ni}^{(\ell)})^T (\mathbf{t}_n - \mathbf{t}_n^{(\ell)}), \quad (67)$$

where $\mathbf{d}_{ni}^{(\ell)} \triangleq \mathbf{t}_n^{(\ell)} - \mathbf{t}_i$. Replacing $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ in (13e) by the right-hand side of (67) yields the linear constraint

$$(\mathbf{d}_{ni}^{(\ell)})^T \mathbf{t}_n \geq b_{ni}^{(\ell)} \triangleq (\mathbf{d}_{ni}^{(\ell)})^T \mathbf{t}_n^{(\ell)} + \frac{D_{\min}^2 - \|\mathbf{d}_{ni}^{(\ell)}\|^2}{2}, \quad (68)$$

Algorithm 2 Two-Timescale AO Algorithm for (P2)

Require: Statistical CSI, P_{tot} , $\mathcal{C}$, $D_{\min}$, δ_0 , and tolerance ε .

- 1: Initialize a feasible $\mathbf{t}^{(0)}$, and obtain $(\mathbf{p}^{(0)}, q^{(0)})$ by Algorithm 1; set $\ell = 0$.
 - 2: **repeat**
 - 3: **for** $n = 1$ to N **do**
 - 4: Compute Ω and $\mathbf{v}$ by (61) and (59), and $\nabla_{\mathbf{t}_n} F$ by (65).
 - 5: Set $\delta_n = \delta_0$.
 - 6: **repeat**
 - 7: Solve (P5.n) to obtain $\mathbf{t}_n^{(\ell+1)}$.
 - 8: **if** $F(\mathbf{t}_n^{(\ell+1)}) < \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)})$ **then** $\delta_n \leftarrow 2\delta_n$.
 - 9: **until** $F(\mathbf{t}_n^{(\ell+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)})$
 - 10: **end for**
 - 11: Obtain $(\mathbf{p}', q')$ by Algorithm 1 with $\mathbf{t}^{(\ell+1)}$.
 - 12: Set $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}', q')$ if this does not decrease F ; otherwise, set $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}^{(\ell)}, q^{(\ell)})$.
 - 13: $\ell \leftarrow \ell + 1$.
 - 14: **until** the fractional increase of F is below ε
- Ensure:** $\mathbf{t}^* = \mathbf{t}^{(\ell)}$, $\mathbf{p}^* = \mathbf{p}^{(\ell)}$, and $q^* = q^{(\ell)}$.
-

whose feasible set is a convex inner approximation of that of (13e).

5) *Subproblem for MA n* : Based on (66) and (68), $\mathbf{t}_n$ is updated by solving

$$(\text{P5.n}) : \max_{\mathbf{t}_n} \hat{F}_n(\mathbf{t}_n | \mathbf{t}_n^{(\ell)}) \quad (69a)$$

$$\text{s.t. } \mathbf{t}_n \in \mathcal{C}, \quad (69b)$$

$$(\mathbf{d}_{ni}^{(\ell)})^T \mathbf{t}_n \geq b_{ni}^{(\ell)}, \forall i \neq n. \quad (69c)$$

By completing the square, (P5.n) is equivalent to the Euclidean projection of $\mathbf{t}_n^{(\ell)} + \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(\ell)}) / \delta_n$ onto the polygon defined by the box constraint and the $N - 1$ linear constraints (69c), i.e., a convex quadratic program (QP) with two variables, which can be solved efficiently by standard solvers (e.g., CVX [16]). In particular, if the point

$$\mathbf{t}_n^{(\ell+1)} = \Pi_{\mathcal{C}} \left(\mathbf{t}_n^{(\ell)} + \frac{1}{\delta_n} \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(\ell)}) \right) \quad (70)$$

satisfies (69c), it is the optimal solution of (P5.n), where $[\Pi_{\mathcal{C}}(\mathbf{x})]_i = \min\{\max\{\mathbf{x}_i, -A/2\}, A/2\}$.

Since $\mathbf{t}_n^{(\ell)}$ is feasible for (P5.n), the update ensures

$$F(\mathbf{t}_n^{(\ell+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell+1)} | \mathbf{t}_n^{(\ell)}) \geq \hat{F}_n(\mathbf{t}_n^{(\ell)} | \mathbf{t}_n^{(\ell)}) = F(\mathbf{t}_n^{(\ell)}), \quad (71)$$

and $\mathbf{t}_n^{(\ell+1)}$ is feasible for (P4) due to (67). Hence, the objective of (P4) is non-decreasing after each element-wise update.

D. Overall Algorithm, Convergence, and Complexity

The overall algorithm is summarized in Algorithm 2. In each iteration, the element-wise position updates do not decrease the objective value of (P2) according to (71), and the power allocation is updated only if the objective value is not decreased.

Hence, the objective value of (P2) is non-decreasing over the iterations. Moreover, it is upper-bounded due to the finite transmit power, e.g., $F \leq \sum_m \log_2(1 + \zeta_m(N - M + \kappa_m N)P_{\text{tot}})$ since $\mu_m \leq \kappa_m N$ and $p_m \leq P_{\text{tot}}$. Therefore, Algorithm 2 is guaranteed to converge.

Complexity: In each iteration, forming Φ and Ω requires $\mathcal{O}(NM^2 + M^3)$ operations. For each MA, Γ_n^{-1} is obtained from Ω by a rank-one update, and the gradient in (65) and each evaluation of F require $\mathcal{O}(M^2)$ operations using (61), while checking or solving (P5.n) involves $\mathcal{O}(N)$ linear constraints. Algorithm 1 requires $\mathcal{O}(|\mathcal{Q}|I_b M)$ operations, where $|\mathcal{Q}|$ is the number of evaluated values of q , including those in the golden-section refinement, and I_b is the number of bisection iterations. Hence, the overall complexity of Algorithm 2 is $\mathcal{O}(I_o(NM^2 + M^3 + NJ(M^2 + N) + |\mathcal{Q}|I_b M))$, where I_o is the number of outer iterations and J is the average number of backtracking trials. Note that the complexity does not depend on the number of channel realizations, in contrast to stochastic optimization methods based on sample averages [17]. Moreover, Algorithm 2 is executed only once per frame, while the small-timescale ZF precoding and AN design in each CI requires $\mathcal{O}(N^2 M + M^3)$ operations.

APPENDIX A: DERIVATIVES OF μ_m , $\bar{c}_m$, AND $\bar{d}$

Let $\omega_m \triangleq [\Omega]_{mm}$, $o_m \triangleq [\Omega^2]_{mm}$, $\tau \triangleq \text{tr}(\Omega)$, and $\tilde{\mathbf{v}} \triangleq \Omega \mathbf{v}$. Since Ω is Hermitian, (64) can be rewritten as $\dot{\Omega} = -(\Omega \dot{\mathbf{k}}_n)(\Omega \mathbf{k}_n)^H - (\Omega \mathbf{k}_n)(\Omega \dot{\mathbf{k}}_n)^H$. Hence, the derivatives of ω_m , o_m , and τ w.r.t. $u \in \{x_n, y_n\}$ are given by

$$\dot{\omega}_m = [\dot{\Omega}]_{mm} = -2\text{Re}\{[\Omega \dot{\mathbf{k}}_n]_m [\Omega \mathbf{k}_n]_m^*\}, \quad (72)$$

$$\begin{aligned} \dot{o}_m &= 2\text{Re}\{[\Omega \dot{\Omega}]_{mm}\} = -2\text{Re}\{[\Omega^2 \dot{\mathbf{k}}_n]_m [\Omega \mathbf{k}_n]_m^* \\ &\quad + [\Omega^2 \mathbf{k}_n]_m [\Omega \dot{\mathbf{k}}_n]_m^*\}, \end{aligned} \quad (73)$$

$$\dot{\tau} = \text{tr}(\dot{\Omega}) = -2\text{Re}\{\mathbf{k}_n^H \Omega^2 \dot{\mathbf{k}}_n\}, \quad (74)$$

where (73) follows from $[\dot{\Omega}\Omega]_{mm} = [\Omega\dot{\Omega}]_{mm}^*$. Then, by differentiating (28), we obtain

$$\begin{aligned} \dot{\mu}_m &= -\frac{\dot{\omega}_m}{\omega_m^2} \left(1 - \tau + \frac{o_m}{\omega_m}\right) \\ &\quad + \left(\frac{1}{\omega_m} - N\right) \left(\frac{\dot{o}_m \omega_m - o_m \dot{\omega}_m}{\omega_m^2} - \dot{\tau}\right). \end{aligned} \quad (75)$$

For $\bar{c}_m$ in (37), the derivative of $\tilde{\mathbf{v}} = \Omega \mathbf{v}$ is

$$\dot{\tilde{\mathbf{v}}} = \dot{\Omega} \mathbf{v} + \Omega \dot{\mathbf{v}} = -\Omega \dot{\mathbf{k}}_n (\mathbf{k}_n^H \tilde{\mathbf{v}}) - \Omega \mathbf{k}_n (\dot{\mathbf{k}}_n^H \tilde{\mathbf{v}}) + \Omega \dot{\mathbf{v}}, \quad (76)$$

and thus

$$\dot{c}_m = \frac{2\text{Re}\{[\tilde{\mathbf{v}}]_m^* [\dot{\tilde{\mathbf{v}}}]_m\} + N \dot{o}_m - \bar{c}_m \dot{\omega}_m}{\omega_m}. \quad (77)$$

For $\bar{d}$ in (38), since $\mathbf{v}^H \dot{\Omega} \mathbf{v} = -2\text{Re}\{(\tilde{\mathbf{v}}^H \dot{\mathbf{k}}_n)(\mathbf{k}_n^H \tilde{\mathbf{v}})\}$, we have

$$\frac{\partial(\mathbf{v}^H \Omega \mathbf{v})}{\partial u} = 2\text{Re}\{\tilde{\mathbf{v}}^H \dot{\tilde{\mathbf{v}}}\} - 2\text{Re}\{(\tilde{\mathbf{v}}^H \dot{\mathbf{k}}_n)(\mathbf{k}_n^H \tilde{\mathbf{v}})\}, \quad (78)$$

$$\dot{d} = -\frac{\partial(\mathbf{v}^H \Omega \mathbf{v})}{\partial u} (1 - \tau) - (N - \mathbf{v}^H \Omega \mathbf{v}) \dot{\tau}. \quad (79)$$

Substituting (75), (77), and (79) into (65) yields $\nabla_{\mathbf{t}_n} F$. All the quantities in (72)–(79) are obtained from matrix-vector products with Ω , which require $\mathcal{O}(M^2)$ operations.

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